

$\Sigma \vec{F}$ + REACTION
+ BODY
+ SURFACE
- NORMAL
- SHEAR

FINAL
VERSION

FLUID MECHANICS

FLUID STATICS

FLUID DYNAMICS

$$\Sigma \vec{F} = 0$$

HYDROSTATIC PRESSURE

$$P_L = P_u + \gamma z$$

$$\gamma = \rho g$$

$$SG = \rho / \rho_w \text{ (SOLID)}$$

RESULTANT FORCE ON FLAT SURFACE

$$F_R = P_c A$$

$$y_p = y_c + \frac{I_{xxc}}{y_c A}$$

BUOYANCY

$$F_B = \Sigma \gamma_i V_i$$

CONSERVATION OF ENERGY

GENERAL

CONSERVATION OF MOMENTUM

$$\frac{d(m\vec{V})}{dt} = \Sigma \vec{F} + \Sigma m_i \vec{V}_i - \Sigma m_e \vec{V}_e$$

$$x: \frac{d(mV_x)}{dt} = \Sigma F_x + \Sigma m_i V_{ix} - \Sigma m_e V_{ex}$$

BERNOULLI

$$\left(\frac{P}{\rho g} + \frac{V^2}{2g} + z \right)_i = \left(\frac{P}{\rho g} + \frac{V^2}{2g} + z \right)_e$$

$$0 = h_g + h_f + h_w + \left(\frac{u}{g} + \frac{P}{\rho g} + \frac{V^2}{2g} + z \right)_i - \left(\frac{u}{g} + \frac{P}{\rho g} + \frac{V^2}{2g} + z \right)_e$$

HEAT TRANSFER

FINAL
VERSION

MODES

CONVECTION

$$\dot{Q} = hA(T_H - T_c)$$

RADIATION

$$\dot{Q} = \epsilon \sigma A (T_H^4 - T_c^4)$$

$$\dot{Q} = h_r A (T_H - T_c)$$

CONDUCTION

$$\dot{Q} = -kA \frac{dT}{dn}$$

CARTESIAN

$$\dot{Q} = \frac{kA(T_H - T_c)}{\Delta x}$$

RADIAL

$$\dot{Q} = \frac{2\pi kL(T_H - T_c)}{\ln(r_o/r_i)}$$

RESISTANCE FORM FOR MULTIMODE HEAT TRANSFER

$$\dot{Q} = \frac{T_H - T_c}{\Sigma R}$$

SERIES $R = \Sigma R_i$

PARALLEL $\frac{1}{R} = \Sigma \frac{1}{R_i}$

RESISTANCES

CONVECTION $R_{conv} = \frac{1}{hA}$

RADIATION $R_{rad} = \frac{1}{h_r A}$

CONDUCTION $R_{cond} = \frac{\Delta x}{kA}$

$$R_{cond} = \frac{\ln(r_o/r_i)}{2\pi kL}$$

SYSTEM

CLOSED

OPEN

TRANSIENT

$$\frac{dm}{dt} = 0$$

$$\frac{dE}{dt} = \dot{Q} - \dot{W}$$

$$\frac{dS}{dt} = \sum \frac{\dot{Q}_j}{T_b} + \dot{\sigma}$$

$$\frac{d\psi}{dt} = \sum \left(1 - \frac{T_0}{T_j}\right) \dot{Q}_j - \left[\dot{W} - P_0 \frac{dV}{dt}\right] - T_0 \dot{\sigma}$$

STEADY-STATE

$$0 = 0$$

$$0 = \dot{Q} - \dot{W}$$

$$0 = \sum \frac{\dot{Q}_j}{T_b} + \dot{\sigma}$$

$$0 = \sum \left(1 - \frac{T_0}{T_j}\right) \dot{Q}_j - \left[\dot{W} - P_0 \frac{dV}{dt}\right] - T_0 \dot{\sigma}$$

NET CHANGE

$$m_2 - m_1 = 0$$

$$E_2 - E_1 = Q - W$$

$$S_2 - S_1 = \sum \frac{Q_j}{T_b} + \sigma$$

$$\psi_2 - \psi_1 = \int_1^2 \left(1 - \frac{T_0}{T_b}\right) \delta Q - [W - P_0(V_2 - V_1)] - T_0 \sigma$$

TRANSIENT

(THE GOVERNING EQUATIONS)
(ALL OTHERS A SIMPLIFIED VERSION)

$$\frac{dm}{dt} = \sum \dot{m}_i - \sum \dot{m}_e$$

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i\right) - \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e\right)$$

$$\frac{dS}{dt} = \sum \frac{\dot{Q}_j}{T_b} + \sum \dot{m}_i s_i - \sum \dot{m}_e s_e + \dot{\sigma}$$

$$\frac{d\psi}{dt} = \sum_j \left(1 - \frac{T_0}{T_j}\right) \dot{Q}_j - [\dot{W} - P_0 \frac{dV}{dt}] + \sum \dot{m}_i \psi_{fi} - \sum \dot{m}_e \psi_{fe} - T_0 \dot{\sigma}$$

STEADY-STATE

$$0 = \sum \dot{m}_i - \sum \dot{m}_e$$

$$0 = \dot{Q} - \dot{W} + \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i\right) - \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e\right)$$

$$0 = \sum \frac{\dot{Q}_j}{T_b} + \sum \dot{m}_i s_i - \sum \dot{m}_e s_e + \dot{\sigma}$$

$$0 = \sum_j \left(1 - \frac{T_0}{T_j}\right) \dot{Q}_j - \dot{W} + \sum \dot{m}_i \psi_{fi} - \sum \dot{m}_e \psi_{fe} - T_0 \dot{\sigma}$$

NET CHANGE (AKA - UNIFORM FLOW UNIFORM STATE)

$$m_2 - m_1 = \sum \dot{m}_i - \sum \dot{m}_e$$

$$E_2 - E_1 = Q - W + \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i\right) - \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e\right)$$

$$S_2 - S_1 = \sum \frac{Q_j}{T_b} + \sum \dot{m}_i s_i - \sum \dot{m}_e s_e + \sigma$$

$$\psi_2 - \psi_1 = \sum_j \left(1 - \frac{T_0}{T_j}\right) Q_j - [W - P_0(V_2 - V_1)] + \sum \dot{m}_i \psi_{fi} - \sum \dot{m}_e \psi_{fe} - T_0 \sigma$$

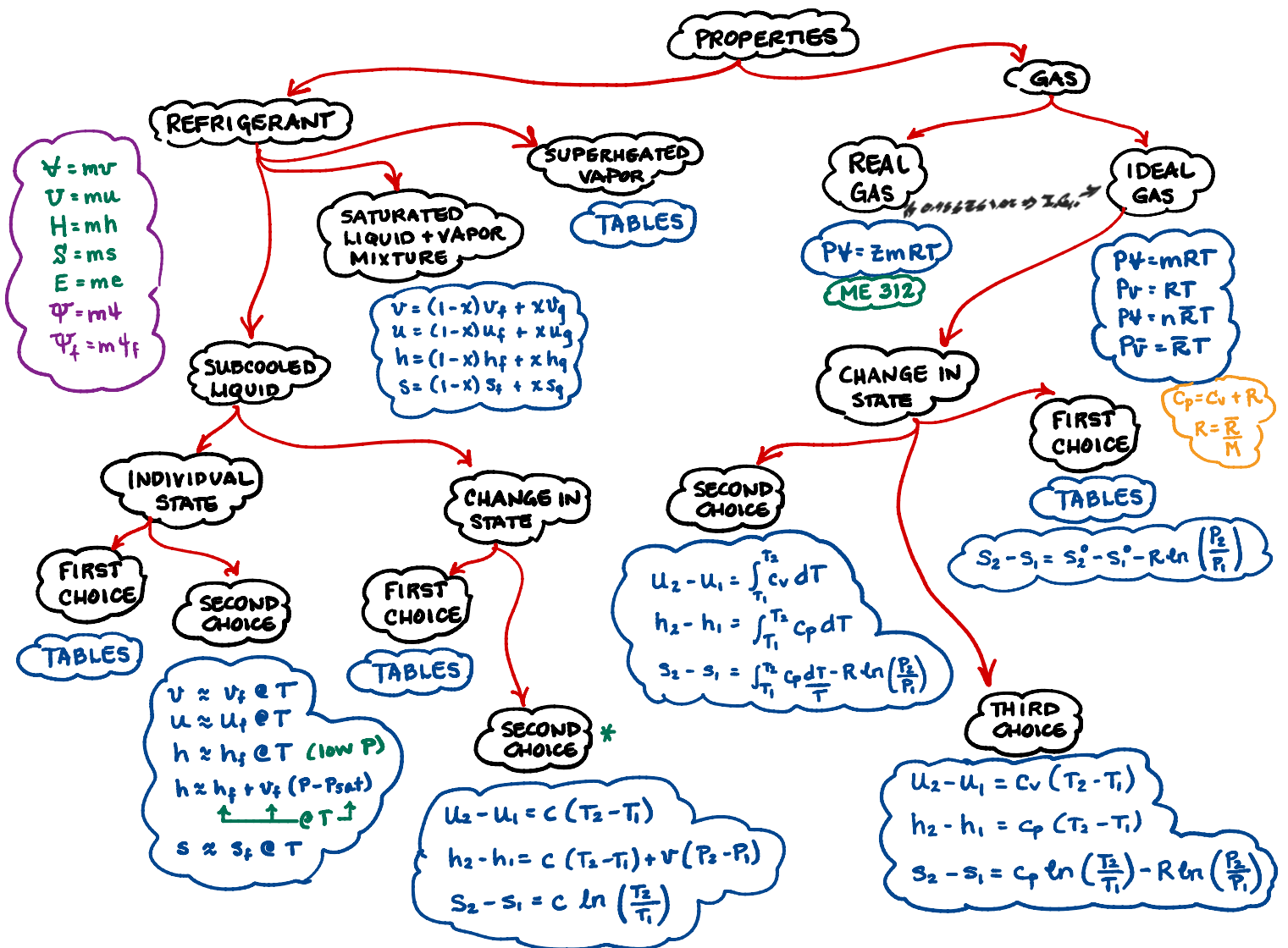
$$\dot{m} = \rho A V = \frac{A V}{v}$$

$$e = u + \frac{V^2}{2} + gz$$

$$h = u + P v$$

$$\psi = (u - u_0) + P_0(v - v_0) - T_0(s - s_0) + \frac{V^2}{2} + gz$$

$$\psi_f = h - h_0 - T_0(s - s_0) + \frac{V^2}{2} + gz$$



* APPLICABLE TO ALL INCOMPRESSIBLE SUBSTANCES

$$W_b = \int_{v_1}^{v_2} P dv$$

PROCESSES

$$\dot{W}_{IR} = -\dot{m} \int_{P_1}^{P_2} v dP$$

REFRIGERANT OR IDEAL GAS

ISOCORIC

$$v_2 = v_1$$

$$W_b = 0$$

ISOBARIC

$$P_2 = P_1$$

$$W_b = P_2 v_2 - P_1 v_1$$

ISOTHERMAL

$$T_2 = T_1$$

$$W_b \text{ (N/A)}$$

POLYTROPIC

$$P_2 v_2^n = P_1 v_1^n$$

$$n=1$$

$$W_b = P_1 v_1 \ln \left(\frac{v_2}{v_1} \right)$$

$$\dot{W}_{IR} = -\dot{m} P_1 v_1 \ln \left(\frac{P_2}{P_1} \right)$$

$$n \neq 1$$

$$W_b = \frac{P_2 v_2 - P_1 v_1}{1-n}$$

$$\dot{W}_{IR} = \dot{m} \left(\frac{n}{1-n} \right) [P_2 v_2 - P_1 v_1]$$

ISENTROPIC

$$S_2 = S_1$$

$$W_b \text{ (N/A)}$$

IDEAL GAS

ISOCORIC

$$W_b = 0$$

$$\frac{T_2}{T_1} = \frac{P_2}{P_1}$$

ISOBARIC

$$W_b = m_2 R T_2 - m_1 R T_1$$

$$\frac{T_2}{T_1} = \frac{v_2}{v_1}$$

ISOTHERMAL

$$W_b = m_1 R T_1 \ln \left(\frac{v_2}{v_1} \right)$$

$$P_2 v_2 = P_1 v_1$$

$$\dot{W}_{IR} = -\dot{m} R T_1 \ln \left(\frac{P_2}{P_1} \right)$$

POLYTROPIC

$$n=1$$

$$W_b = m_1 R T_1 \ln \left(\frac{v_2}{v_1} \right)$$

$$\dot{W}_{IR} = -\dot{m} R T_1 \ln \left(\frac{P_2}{P_1} \right)$$

$$n \neq 1$$

$$W_b = \frac{m_2 R T_2 - m_1 R T_1}{1-n}$$

$$\dot{W}_{IR} = \dot{m} \left(\frac{n}{1-n} \right) R (T_2 - T_1)$$

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2} \right)^{n-1}$$

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{(1-n)/n}$$

ISENTROPIC

$$W_b = \frac{m_2 R T_2 - m_1 R T_1}{1-k}$$

$$\dot{W}_{IR} = \dot{m} \left(\frac{k}{1-k} \right) R (T_2 - T_1)$$

CONSTANT
SPECIFIC
HEAT

TABLES

$$P_2 v_2^k = P_1 v_1^k$$

$$\frac{T_{2s}}{T_1} = \left(\frac{v_1}{v_2} \right)^{k-1}$$

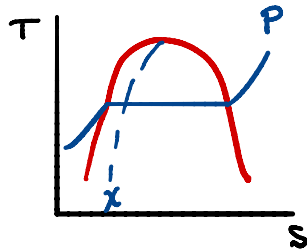
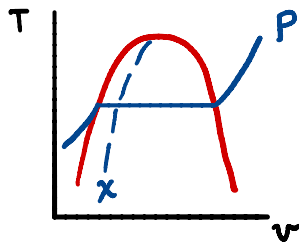
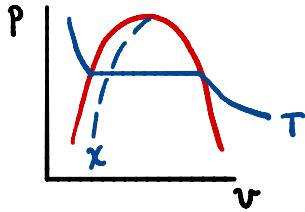
$$\frac{T_{2s}}{T_1} = \left(\frac{P_2}{P_1} \right)^{(1-k)/k}$$

$$\frac{P_2}{P_1} = \frac{P_{r2,s}}{P_{r1}}$$

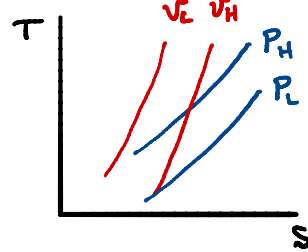
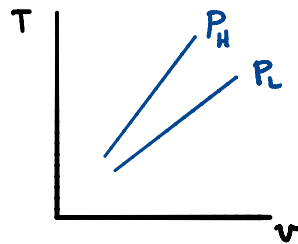
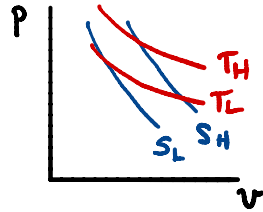
$$\frac{v_2}{v_1} = \frac{v_{r2,s}}{v_{r1}}$$

DIAGRAMS

REFRIGERANTS



IDEAL GAS



CYCLES

POWER

$$\dot{Q}_H(\text{IN})(+)$$

$$\eta = \frac{|\dot{W}_{\text{NET}}|}{|\dot{Q}_H|} = \frac{|\dot{Q}_H| - |\dot{Q}_L|}{|\dot{Q}_H|}$$

$$\eta_{\text{MAX}} = \frac{T_H - T_L}{T_H}$$

REFRIGERATION

$$\dot{Q}_L(\text{IN})(+)$$

$$\beta = \frac{|\dot{Q}_L|}{|\dot{W}_{\text{NET}}|} = \frac{|\dot{Q}_L|}{|\dot{Q}_H| - |\dot{Q}_L|}$$

$$\beta_{\text{MAX}} = \frac{T_L}{T_H - T_L}$$

HEAT PUMP

$$\dot{Q}_L(\text{IN})(+)$$

$$\gamma = \frac{|\dot{Q}_H|}{|\dot{W}_{\text{NET}}|} = \frac{|\dot{Q}_H|}{|\dot{Q}_H| - |\dot{Q}_L|}$$

$$\gamma_{\text{MAX}} = \frac{T_H}{T_H - T_L}$$

$$\text{MEP} = \frac{W_{\text{NET}}}{v_{\text{max}} - v_{\text{min}}}$$

$$r_c = \frac{v_{\text{END}}}{v_{\text{START}}} \quad (\text{ISOBARIC PROCESS})$$

$$r = \frac{v_{\text{max}}}{v_{\text{min}}}$$

$$r_p = \frac{P_{\text{MAX}}}{P_{\text{MIN}}}$$

